

Volume

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS **Volume** is the amount of space inside a solid. You can measure it by filling the solid with **unit cubes** and counting them. For a rectangular prism, volume = length $\times$ width $\times$ height. A prism that is 4 units long, 2 units wide, and 3 units high holds $4 \times 2 \times 3 = \mathbf{24 \text{ cubic units}}$. Volume is always measured in **cubic units** because it counts cubes that fill space.

Round 1 — Retrieve It

1. What does volume measure?

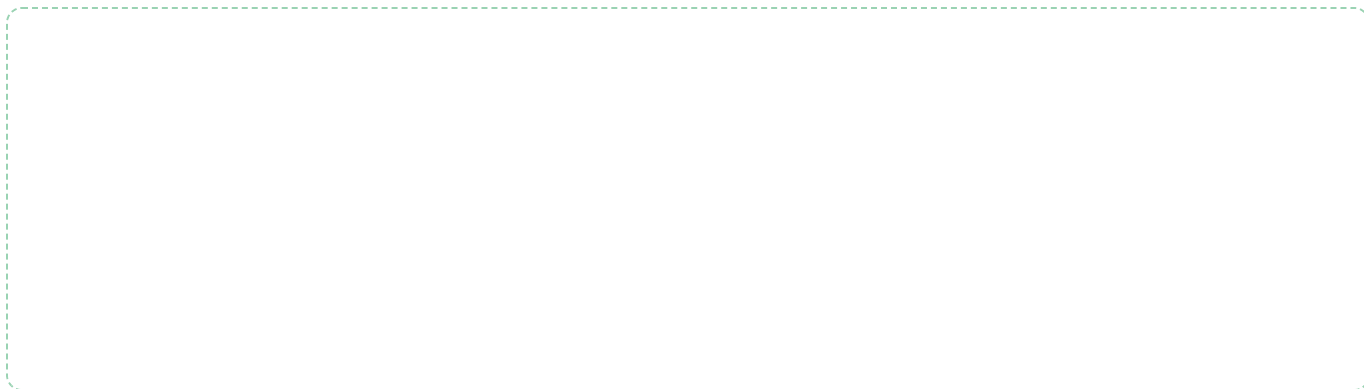
2. What shape do we use to fill a solid and count its volume?

3. What is the formula for the volume of a rectangular prism?

4. What unit is used to measure volume?

Round 2 — Sketch It

Draw a rectangular prism that is 4 units long, 2 units wide, and 3 units high. Draw unit cubes inside one bottom layer to show how it is filled.



Volume — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. A box is 4 units long, 2 units wide, and 3 units high. Find its volume. Show the multiplication.

2. A cube is 3 units on every side. Find its volume.

3. A prism has a bottom layer of 10 cubes and is 4 layers tall. How many unit cubes fill it?

Round 4 — Reason from Evidence

A friend counted only the cubes on the outside of a prism and said that was the volume. Explain why counting only the outside cubes gives the wrong volume.

Volume — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain how filling a prism with unit cubes is the same as multiplying length $\times$ width $\times$ height.

C**Connect**

Connect volume to a real object you could fill, such as a box, a fish tank, or a moving crate. Tell what you would be measuring.

E**Extend**

Describe two different prisms that both have a volume of 24 cubic units. Give the length, width, and height of each.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Volume

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. The amount of space inside a solid.
2. Unit cubes.
3. Volume = length $\times$ width $\times$ height.
4. Cubic units.

Sketch It — look for

A rectangular prism labeled 4 by 2 by 3 with a bottom layer showing 8 unit cubes (4×2) drawn inside.

Apply It

1. Volume = $4 \times 2 \times 3 = 24$ cubic units.
2. Volume = $3 \times 3 \times 3 = 27$ cubic units.
3. $10 \times 4 = 40$ unit cubes.

Reason from Evidence — look for

Volume counts every cube that fills the whole space, including the cubes hidden inside. Counting only the outside cubes leaves out the inside ones, so the total is too low.

ACE — Extend look-for

Any two of: $4 \times 2 \times 3$, $2 \times 2 \times 6$, $1 \times 4 \times 6$, or $1 \times 1 \times 24$ — each gives 24 cubic units.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.